



# Project Close-out Report

(Dead-man-switch) Permission based Wallet, shipped as Epora

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| <b>Name of project</b>        | (Dead-man-switch) Permission based Wallet, shipped as Epora                                       |
| <b>Project URL</b>            | <a href="https://projectcatalyst.io">projectcatalyst.io</a> , Fund 11, Cardano Use Cases: Concept |
| <b>Project number</b>         | 1100002                                                                                           |
| <b>Project manager</b>        | Sandro Schaier                                                                                    |
| <b>Date project started</b>   | 11 March 2024                                                                                     |
| <b>Date project completed</b> | 26 September 2026                                                                                 |
| <b>Plan</b>                   | 7 months, with deliveries from April to October 2024                                              |
| <b>Budget</b>                 | 100,000 ADA in five milestones                                                                    |
| <b>Close-out video</b>        | <a href="#">Watch the close-out video on YouTube</a>                                              |

## 1 Challenge KPIs and how the project addressed them

The challenge brief states aims, not numeric KPIs, so each row quotes one aim and the project's answer.

| Aim in the challenge brief                                                                  | How the project addressed it                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "deliver proof of concept, design research and basic prototyping"                           | The whitepaper v1.0 gives the design, a formal model, and a security analysis. The Aiken validators and the reference app run on Preprod at <a href="https://epora.io">epora.io</a> . Mainnet beta: <a href="https://mainnet.epora.io">mainnet.epora.io</a> . |
| "generate and test new ideas that bring utility to Cardano"                                 | Epora brings the smart-contract wallets of large treasuries to individual users. Recovery, shared control, allowances, and streaming payments combine as data on one contract. 750 contract tests and a public Preprod phase tested the ideas.                |
| "produce specifications and a pathway that requires more significant technical development" | The whitepaper and the OpenAPI 3.1 document specify the system. Section 5 gives the pathway: a smart-wallet CIP, an external audit, and wallet integrations.                                                                                                  |
| "for the benefit of the Cardano ecosystem"                                                  | The project's own code and documents are public under the MIT license. Third-party parts keep their own licenses.                                                                                                                                             |

## 2 Project KPIs and how the project addressed them

The proposal measures success through its five milestones. The table summarizes the acceptance criteria from the Statement of Milestones, the result, and the review status in the Milestone Module on 25 September 2026.

| Milestone                          | Acceptance criteria                                                                                                                             | Result                                                                                                                                                                                                     | Status                  |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1. Setup, whitepaper, planning     | X and Discord accounts; a whitepaper (abstract, introduction, functions, permission system, use cases); a task list                             | X account <a href="https://twitter.com/schaier_io">@schaier_io</a> and a Discord server; <a href="#">whitepaper v1.0</a> with its LaTeX source; a <a href="#">task tracker</a> with one file per milestone | Approved<br>7 Jul 2026  |
| 2. Smart contract development      | The whitepaper methods; code documentation for new developers; abuse vectors explained and shown not to work                                    | Aiken validators for every whitepaper transition; a contracts guide; a <a href="#">security evidence map</a> ; 750 of 750 contract tests pass                                                              | Approved<br>6 Aug 2026  |
| 3. Development of UI               | A video of the whitepaper functions; an API and a specification for external developers; code documentation                                     | A wallet UI video; a public v1 API with an OpenAPI 3.1 document and a <a href="#">developer guide</a> ; the app README                                                                                     | Approved<br>21 Sep 2026 |
| 4. Testnet launch                  | A usable prototype; aggregated feedback; improvements and bug fixes from the feedback                                                           | The prototype on Preprod at <a href="https://epora.io">epora.io</a> ; a <a href="#">feedback page</a> with 49 categorized issues; 46 fixes with linked pull requests                                       | Submitted,<br>in review |
| 5. Finalization and mainnet launch | A mainnet prototype; categorized feedback; talks with potential wallet developers, if they or the community ask; this report; a close-out video | Mainnet beta at <a href="https://mainnet.epora.io">mainnet.epora.io</a> ; the same feedback page; a <a href="#">wallet outreach page</a> ; this report; the video in Section 8                             | This<br>submission      |

**Follow-up.** The Vesper retest is complete (*reported*). Wallet outreach is complete. Integration talks continue.

**Timeline and scope.** The plan had seven months, with the last delivery in October 2024. The Milestone Module approved milestones 1 to 3 between 7 July and 21 September 2026. The project delivered much more than the proposal asked for, and the budget stayed at 100,000 ADA. The larger scope took more time than the plan allowed.

| Proposal                                                                                       | Delivered                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Permissions after a period of inactivity<br>Two parties agree before a transaction             | A proof-of-life dead-man-switch that owners and liveness keepers renew<br>Weighted multi-signature for operator actions, with a workspace where co-signers review and sign                                                                                                               |
| Limits for spending certain assets<br>A family member recovers funds after six inactive months | A per-day allowance for each user, with up to 5 assets in each allowance<br>Weighted-share recovery for several beneficiaries, each with an optional unlock delay, and exact distribution to all of them                                                                                 |
| Partially custodial wallets                                                                    | Third parties can hold a role: co-owner, spender, liveness keeper, or beneficiary. The funds stay in the contract.                                                                                                                                                                       |
| Not in the proposal                                                                            | Up to 15 streaming payments that settle while the owner is offline; reward withdrawal, certificates, and governance votes from the smart wallet;<br>stake-credential pinning and consolidation; a formal model and a security analysis;<br>750 contract tests, 37 of them property-based |

### 3 Key achievements

- **Specification.** The whitepaper v1.0 specifies the permission system with a formal model, 11 theorems, and a security analysis.
- **Contracts with security evidence.** Aiken validators implement every whitepaper transition, and 750 contract tests pass. A security evidence map links each abuse vector to the tests that reject it.
- **App and public API.** The web app creates, configures, and uses smart wallets on Preprod. The public API builds unsigned transactions that any wallet can sign, and the server never holds a key.
- **Community testing.** Reports came from team test sessions, GitHub issue forms linked in the app, and Discord. All 49 tracked issues have one category, and 46 are fixed. Reports from two outside testers led to changes ([#562](#), [#568](#) to [#571](#)).
- **Mainnet beta.** Network configuration, beta consent, legal pages, and a network switch are merged ([#589](#), [#590](#), [#594](#)). The beta runs at [mainnet.epora.io](https://mainnet.epora.io).
- **Wallet developer outreach.** We contacted wallet developers in a public channel and by direct message (*reported*). We asked about a native integration and a CIP for a standard smart wallet. One team replied (*reported*).

### 4 Key learnings



One action in Epora: two contracts read the same redeemer, and a normal wallet signs and pays.

- **Two contracts must agree on one action.** Both contracts read one redeemer, the declared action. The state contract proves that the action equals the true change to the wallet state. The wallet contract limits the payout to that action and has no permission logic, so a reviewer can check it by itself.
- **One configurable contract replaces many bespoke ones.** Six features give 364 combinations of “and” and “or”, and seven give 1,093. One contract for each combination is too much to build, test, and audit. So the whole configuration is one datum on one contract, and a new combination is data, not code.
- **An open deposit address has edge cases.** Anyone can pay into the wallet with no datum, as into a normal address. But a sender can attach a datum hash, or pick a foreign stake credential that then collects the staking rewards of those funds. The contracts now pin the stake credential and reject datum hashes on outputs that stay in the wallet. An earlier whitepaper missed the datum-hash case and now records the correction.

- **Script size needs its own design work.** A transaction may use at most 16,384 bytes, and the state script alone has 14,469 bytes. So the script is stored once as a reference script, in a 93-byte always-fail store at a fixed address. Each state transaction points to it. Without the reference, even the mint of a new wallet would exceed the limit. With both scripts as references, a payout to 15 beneficiaries used 9,925 signed bytes. With the wallet script inline, 2 beneficiaries used 11,163. With full traces the state script has 40,258 bytes, so the production build drops them. A size pass used standard-library helpers, callbacks, and shared checks (#296). It cut the state script from 15,936 to 14,428 bytes and the wallet script from 9,810 to 9,378 bytes.
- **Caps and gates keep each action inside the limits.** We try to prove that every action fits in one transaction, so that no wallet state can lock the funds forever. The contracts cap the lists: 15 user and beneficiary records (at most 10 users), 15 streaming payments, and 5 assets in one allowance. The beneficiary checks compare every pair, so their cost grows with the square of the list. Budget tests run actions on wallets at the caps. CI fails when a recorded cost moves by more than 1% or passes a fixed ceiling. The app checks each signed transaction against the size limit before it submits. Each configuration must also keep a recovery path: an admin, a quorum, or a beneficiary that can still sign.
- **Two wallets work together by design.** The smart wallet holds the funds and enforces the rules. The normal wallet holds the user's keys, signs each action, pays the fee, and provides the collateral. So the app builds each action as one transaction that uses both wallets. The CIP in Section 5 aims to integrate the two wallets. The normal wallet will likely stay part of the setup.

## 5 Sustainability and next steps

Epora continues as an open-source project. Received Catalyst funds cover servers and other operating costs. Our team plans to maintain the beta and address feedback through the public issue tracker. Longer-term, a smart-wallet CIP could support shared maintenance with wallet teams. Future funding or revenue would depend on adoption and partner agreements.

- **Mainnet beta.** Run the beta with small funds and keep its permanent unaudited-code notice.
- **A smart-wallet CIP.** EVM chains have smart-account standards, for example ERC-4337 and ERC-7579. We plan to draft a similar CIP for Cardano. It would define how a wallet finds a smart wallet, shows its funds and rules, and builds its actions. It can build on CIP-141 (Plutus wallets) and CIP-106 (multisig wallets), which both have the status Proposed. After an external audit, the Epora contracts could be the reference implementation.
- **Audit funding.** We plan to fund the external audit after the community gives feedback on the CIP, if the audit is needed. This depends on the community and on the CIP implementations.
- **Wallet integrations.** Follow up with the wallet teams that we contacted. At TOKEN2049, we plan to meet builders in person and collect their feedback.

## 6 Final thoughts

The proposal addressed the risk of losing all funds when a key or a backup is lost. The delivered contracts, app, and API give Cardano users recovery and shared permissions without a custodian of funds. We intend this work to become the base of a smart-wallet standard for Cardano.

## 7 Links to other project sources

|                               |                                                                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Documents</b>              | <a href="#">Repository</a> ; <a href="#">whitepaper</a> ; <a href="#">security evidence map</a> ; <a href="#">API guide</a> ; <a href="#">live API reference</a> |
| <b>Progress</b>               | <a href="#">Task tracker</a> ; <a href="#">categorized feedback</a> ; <a href="#">wallet outreach</a>                                                            |
| <b>Prototypes</b>             | Preprod at <a href="#">epora.io</a> ; mainnet beta at <a href="#">mainnet.epora.io</a>                                                                           |
| <b>Catalyst and community</b> | <a href="#">Milestone Module</a> ; Discord <a href="#">discord.gg/2uh4BynQBW</a> ; X <a href="#">@schaier_io</a>                                                 |

## 8 Close-out video

Watch the Epora Catalyst Fund 11 close-out video on YouTube.

**Sources.** Catalyst project page and Milestone Module (read on 25 September 2026); linked repository files; aiken check, Aiken v1.1.23 (750 passed, 0 failed), and the repository's size checks, run the same day. Items marked (*reported*) have no public record.